METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

DATASHEET R-201

R-201 · Methanol quench converter

DOCUMENT NO.	:	MEOH-V1-ME-DS-001
DOCUMENT TITLE	:	Datasheet R-201
PROJECT	:	Methanol Converter Ver 1 · OTS
STATUS	:	Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

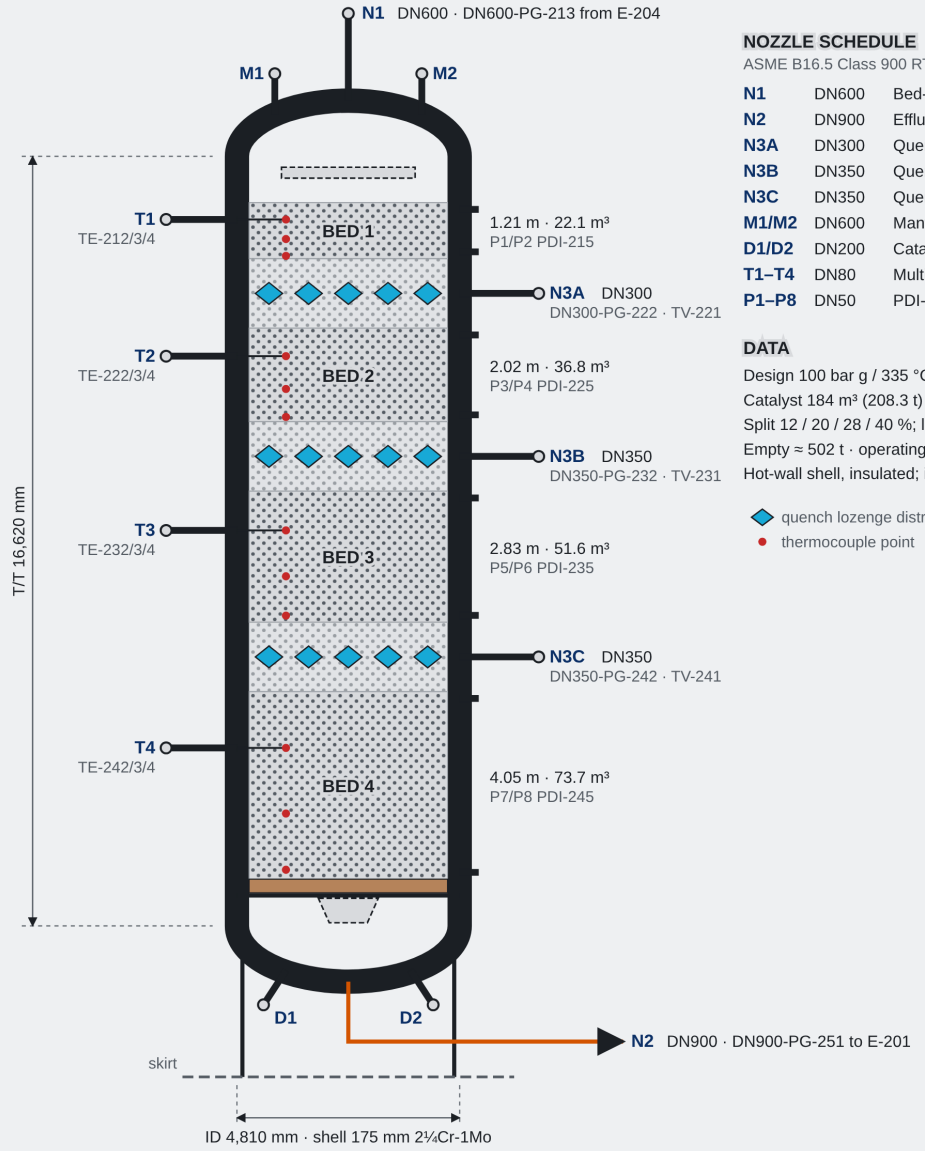
R-201 · Methanol quench converter

Equipment list MEOH-V1-ME-EQL-001. Values from the simulator model at its design point unless marked assumed. P&ID: MEOH-PID-001.

General		
Service	Methanol synthesis, 4 adiabatic Cu/ZnO/Al_2O_3 beds with cold-shot quench	PFD-201, MEOH-PID-001
Type	Vertical, hot-wall, single catalyst mass with lozenge quench distributors	Design tab §2
Quantity	1	
Operating (model, design point)		
Inlet pressure / bed pressures	79.7 bar g / 79.7 · 79.4 · 78.9 · 78.3 bar g	model P.Pin, P.Pbed
Feed (FI-201)	1928 kNm³/h, M 6.57 kg/kmol	FIC-201 SP 1928
Bed inlet / exit temperatures	220/260 · 225/277 · 230/270 · 235/268 °C	TIC-211…241, TI-214…244
Bed ΔP	0.31 · 0.48 · 0.66 · 0.88 bar	PDI-215…245
Heat of reaction	84.6 MW	XI-260
H_2 partial pressure (feed, at design P)	80.8 bar a	y(H_2) 80 %
Design		
Design pressure	100 bar g	Zone B (C-201 discharge side), EQL §2
Design temperature	335 °C	TV-211 E-201 bypass to 100 % (TIC-211 MAN) (301.8 °C) + 30 °C, rounded up to 5
MDMT	−29 °C	assumed; blowdown study TBC (EQ-em-05)
Code	ASME VIII Div. 1 (assumed); TEMA R for exchangers	
Joint efficiency / RT	1.0 / 100 % RT + UT	assumed
Corrosion allowance	3 mm	assumed
PWHT	Yes (Cr-Mo, thickness)	
Hydrotest	130.9 bar g (1.3 × MAWP)	UG-99(b)
Materials		
Shell, heads	2¼Cr-1Mo SA-387 Gr 22 Cl 2	Nelson curve, EQL §3
Forgings / nozzles	SA-336 F22 Cl 3	
Internals (distributors, grid, collector)	SS 321	typical
Catalyst support	Grid + graded alumina balls	Design tab §2
Dimensions and weights		
Inside diameter	4810 mm	Design tab Ergun sizing 4.81 m
Tangent-to-tangent	16620 mm	Σ beds + 3 quench zones 1.5 m + 2.0 m ends
Bed heights 1–4	1.21 · 2.02 · 2.83 · 4.05 m	Design tab
Bed volumes 1–4	22.1 · 36.8 · 51.6 · 73.7 m³	total 184.1 m³ = P&ID 184 m³ (208.3 t, bulk density 1131 kg/m³)
Shell thickness (required + CA → nominal)	170.8 + 3 → 175 mm	UG-27(c)(1)
Head thickness (2:1 ellipsoidal)	164.9 + 3 → 170 mm	UG-32(d)
Weight empty / operating	≈ 502 t / ≈ 725 t	pressure parts +10 %, internals 30 t, balls 15 t (assumed)
Nozzles (ASME B16.5 Class 900 RTJ, assumed)		
N1	DN600	Bed-1 feed inlet (from E-204) · DN600-PG-213
N2	DN900	Effluent outlet (to E-201) · DN900-PG-251
N3A	DN300	Quench inlet bed 2 (TV-221) · DN300-PG-222
N3B	DN350	Quench inlet bed 3 (TV-231) · DN350-PG-232
N3C	DN350	Quench inlet bed 4 (TV-241) · DN350-PG-242
M1/M2	DN600	Manways / catalyst loading (top)
D1/D2	DN200	Catalyst dump (bottom)
T1–T4	DN80	Multipoint thermowells TE-212…244
P1–P8	DN50	PDI-215…245 taps

Notes		
1	Relief: no PSV on R-201; protected by PSV-201 on V-201 (no isolation valves between C-201 discharge and V-201; car-seal any that are added).	MEOH-PID-001 note 6
2	Shell skin thermocouples recommended (HAZOP R06).	Design tab §2
3	Div. 1 gives a heavy wall; Div. 2 (higher allowable stress) would reduce it — vendor option.	

General arrangement sketch (not to construction; dimensions from this datasheet)



NOZZLE SCHEDULE

ASME B16.5 Class 900 RTJ (assumed)

N1	DN600	Bed-1 feed inlet (from E-204)
N2	DN900	Effluent outlet (to E-201)
N3A	DN300	Quench inlet bed 2 (TV-221)
N3B	DN350	Quench inlet bed 3 (TV-231)
N3C	DN350	Quench inlet bed 4 (TV-241)
M1/M2	DN600	Manways / catalyst loading (top)
D1/D2	DN200	Catalyst dump (bottom)
T1–T4	DN80	Multipoint thermowells TE-212…244
P1–P8	DN50	PDI-215…245 taps

DATA

Design 100 bar g / 335 °C · MDMT −29 °C
Catalyst 184 m³ (208.3 t) in 4 beds
Split 12 / 20 / 28 / 40 %; lozenge quench rows
Empty ≈ 502 t · operating ≈ 725 t (assumed)
Hot-wall shell, insulated; internals SS 321

◆ quench lozenge distributor
● thermocouple point